

Getting Started with

GNU Octave

Part 1



A lot of what GNU Octave provides revolves around manipulating matrices, so let's begin our journey with them.

GNU Octave is a software tool for numerical computations. Octave has extensive tools to solve common numerical linear algebra problems, find the roots of non-linear equations, integrate ordinary functions, manipulate polynomials, and integrate ordinary differential and differential-algebraic equations. Octave is available for use on UNIX systems (Linux, Solaris and Mac OSX) and Windows.

If your study or play involves numerical computations, Octave is for you. It has been designed with MATLAB compatibility in mind, so with careful design you could write scripts that run on both MATLAB and Octave. It is easily extensible and customisable via user-defined functions written in Octave's own language, or using dynamically loaded modules written in C++, C, Fortran, or other languages. As we go ahead, we

will look at how we can carry out these functions and more, in Octave. For now, let's start using Octave. The first couple of articles in this series will deal with using Octave for matrix manipulation and linear algebra, in general. I assume readers have some basic knowledge of the mathematics of matrices and linear algebra, and I will not make any effort to explain that.

Installation

As always, you have two options: to use the package manager of your Linux distribution, or compile it from sources. It is always easier to do an *apt-get install* or its other cousins. Please find the name of the Octave package for your own distro.

Let's play with Octave

Fire up the terminal and type `$ octave` and you should see something like the following: